

ERTH 467 – Time Series Analysis and 1D Signal Processing (F2026) Lab Assignment #1

In this lab assignment, you will develop a code to perform forward modelling of reflections, assuming a plane wave in a 1D subsurface model by computing a reflectivity series and convolving this with a source wavelet. The data for this assignment is provided in the files “**erth_467_lab_1_data.mat**” and “**erth_467_lab_1_data.txt**” on D2L (the two files contain the same data; use the **.mat** file if you are coding in Matlab and the **.txt** file otherwise). Based on your results, you will write a brief scientific report (see report format and rubric under Lab Information on D2L) discussing your results. You will submit the assignment to the Dropbox on D2L as a compressed (**.zip**) file entitled “**ERTH_467_Lab_1_Name.zip**” containing a copy of your report in **.pdf** format and a copy of all code used to generate your report results. In the file name, replace **Name** with your surname.

The data file contains both density **rho** (in kg/m³) data and p-wave velocity **vp** (in m/s) data from a well log over a range of depths **z** up to a depth of 3000 m. In the **.txt** file, the first column gives the depth of the top of each layer (in m), the second column gives the density of the layer, and the third column gives the velocity of the layer. Read the data into your program. In MATLAB, you can use the **load()** function to open the data file, and the variables will have the appropriate names.

Define a source wavelet as a single-period sine wave with a frequency of 32 Hz and unit amplitude sampled at the same time step **dt** = 0.001 s.

Create a new **figure()** and use the **subplot()** function to define a grid of plots with one row and four columns. In Python, you can do this by using **import matplotlib.pyplot as plt** and then using the **plt.subplots()** function. Plot the density, p-wave velocity, impedance and reflection coefficients against depth with depth increasing down the vertical axis. Make sure to include meaningful labels on the axes of the plot. Based on these plots, comment in your report on the features of the subsurface stratigraphy.

Write a function to perform the convolution of two vectors with the interface **[w]=conv_467(u,v)** (or **def con_467(u, v):** in Python) where **w** is the output of the convolution. Use any of the approaches discussed in the course notes to perform the convolution of discrete vectors. Make sure that **length(w)=length(u)+length(v)-1**. Verify that your function gives the same results as the MATLAB built-in function **conv()**.

For the following questions, use the properties in the data file to determine reflection coefficients, travel times, etc.

Defining **r_R** as the vector of reflection coefficients corresponding to each two-way travel time (sampled at **dt** = 0.001 s), create a synthetic seismogram defined as **s_R = r_R * w**. Make a plot of **s_R** and **r_R**.

Defining $\mathbf{r}_P$ as the vector of primary reflection amplitudes (with polarity, i.e.

$$r_N = R_N * \Sigma (1 - R_m^2)$$

) corresponding to each two-way travel time, create a synthetic seismogram defined as $\mathbf{s}_P = \mathbf{r}_P * \mathbf{w}$. Make a plot of $\mathbf{s}_P$ and $\mathbf{r}_P$.

Defining $\mathbf{r}_M$ as the vector defining the measured reflections relative to the downgoing wave for both primaries and first-order (three-bounce) multiples corresponding to each two-way travel time, create a synthetic seismogram defined as $\mathbf{s}_M = \mathbf{r}_M * \mathbf{w}$. Make a plot of $\mathbf{s}_M$ and $\mathbf{r}_M$. Hint: Each first-order multiple reflecting first from the layer n , must then experience a downward reflection from layer $m < n$, then a second upgoing reflection from layer $p > m$.

The fourth column of the data file defines a Q value, characterizing attenuation, for each range of depths. We will assume the amplitude loss to attenuation for each layer is

$$e^{-\omega t/(2Q)},$$

where $\omega/(2\pi)$ is the 32 Hz dominant frequency of our wavelet, and t is the time spent propagating through that layer. Defining $\mathbf{r}_Q$ as the vector defining the measured reflections relative to the downgoing wave for both primaries and first-order (two-bounce) multiples corresponding to each two-way travel time (now accounting for attenuation), create a synthetic seismogram defined as $\mathbf{s}_Q = \mathbf{r}_Q * \mathbf{w}$. Make a plot of $\mathbf{s}_Q$ and $\mathbf{r}_Q$.

Write a brief report on your results following the format posted on D2L. The report and code (together in a compressed file) should be submitted to the Dropbox on D2L by 23:30 on Friday, October 17, 2026.